SHABAA ALAM

Aeronautical Engineering Student | Imperial College London

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EDUCATION

Imperial College London — MEng Aeronautical Engineering (year 3 of 4) Expected May 2028

- **Relevant Coursework:** **Aerodynamics**, **Flight Dynamics**, **Structural Analysis**, **Control Systems**, **Advanced Manufacturing Processes**.

TECHNICAL PROJECTS

ICLR Ember Rocket Team (EuRoC) | *Design & Manufacturing Engineer* 2025 – Present

- Designed full **CAD** assemblies for a custom filament winder in **Fusion 360**, adhering strictly to **ISO tolerances**.
- Precision-machined drive shaft components using **manual lathes** to achieve bearing tolerances within **5 microns**.

Duocopter Flight Controller | *2nd Year Systems & Control Project* 2026

- Modelled and simulated a discrete **PID controller** using **MATLAB & Simulink** to stabilize a dual-rotor **UAV**.
- Optimized control parameters for a 43g dynamic structure, reducing Mean Squared Deviation (MSD) to 0.0062 m^2 while limiting total energy usage to 10.38 kJ over a 70-second target altitude profile.

Imperial Formula Racing (Formula Student) | *Composites Engineer* 2024 – 2025

- Engineered an ergonomic **carbon fibre** driver seat inlay using expanding foam, ensuring compliance with competition safety regulations.
- Conducted **risk assessments** and structural safety checks prior to vehicle component manufacturing.

WORK EXPERIENCE & LEADERSHIP

Private & Academy Tutor (A-Level STEM) | *Westcliff Academy for Learning* 2023 – Present

- Communicate complex mathematical and physical concepts in A-Level Maths, Further Maths, and Physics to over 10 students weekly.
- Manage scheduling, client progress reporting, and customized lesson planning alongside a demanding full-time engineering degree.

Teaching Intern | *Westcliff High School for Boys* Jun 2025 – Jul 2025

- Supported delivery of the secondary STEM curriculum and mentored students in structured, analytical problem-solving techniques.

TECHNICAL SKILLS & COMPETENCIES

Coding & Languages:	**C++**, **Python**, **MATLAB**
Analysis & CFD/FEA:	**Abaqus (FEA)**, **STAR-CCM+ (CFD)**, **Simulink**
CAD & Design:	**SolidWorks**, **Fusion 360**, **Dassault 3DEXPERIENCE**
Manufacturing:	**Manual Lathes & Milling**, **Aluminium Casting**, **3D Printing**, **Laser Cutting**
Spoken Languages:	English (Native), Bengali (Fluent)